PROJECT WORK SAMPLE | PREPARED FOR PROFESSIONAL REVIEW
HydroCode Water Evidence and Decisions
System design and a working virtual demonstrator
Developed by Dana Obeidat

Design challenge:

How can a water infrastructure team turn fragmented measurements and operational records into a traceable recommendation, with clear verification requirements and responsibility for action?

My documented contribution:

I developed the HydroCode concept and supporting system architecture, measurement assurance requirements, evidence and escalation workflows, integration specifications, pilot validation protocols and implementation planning materials.

Selected work product structure:

Design component	Purpose and selected elements
Measurement assurance	Source identity, calibration status, data quality flags and missing evidence.
Evidence and review	Connect observations to supporting records, uncertainty and required checks.
Decision workflow	Watch / Verify / Intervene / Escalate, with human review and responsible owners.
Pilot evaluation	Baseline indicators, reproducible alerts, verification findings and acceptance gates.

HydroCode

Prototype and recorded testing:

A virtual demonstrator was deployed on Vercel in August 2026, with a Supabase connection recorded during testing. Development tests covered synthetic pressure and flow readings and navigation across seven interface sections. Additional local tests covered baseline, water quality, energy drift and pressure-drop scenarios; deployment of those additions is not established.
Demonstrator: https://hydrocode-virtual-demonstrator.vercel.app

Relevance to an employer:

This work demonstrates project definition, multidisciplinary requirements coordination, operational workflow design, governance documentation and planning for structured technical review.

Evidence boundary:

The demonstrator uses synthetic data and implements selected functions. It does not establish field deployment, validated predictive accuracy, certified integrations, measured savings or production readiness. The broader four-phase architecture remains subject to implementation and site-specific validation.

Dana Obeidat | https://www.linkedin.com/in/dana-tariq-obeidat/